



BIS 514 – SYSTEMS ANALYSIS, MODELING AND DESIGN

REMOTE CUSTOMER ONBOARDING SYSTEM

Berfin Erin

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1. Introduction

1.1 System Information

Our system name is Remote Customer Onboarding System. Remote Customer Onboarding System is an online customer acquisition system that enables individuals to become bank customers remotely without visiting a physical branch. The system is designed to provide a fast, secure, and user-friendly registration process through mobile or web channels.

In this system, a user starts the application by entering personal information and verifying identity through several control steps such as SMS verification, identity card validation, MRZ/NFC reading, selfie verification, and video call confirmation. After successful verification, the system collects necessary customer information, allows product selection, creates the customer record, and completes the activation and contract delivery steps.

The main purpose of the system is to digitalize the customer onboarding journey, reduce manual workload, shorten processing time, improve customer experience, and ensure compliance with legal and banking regulations.

2. Requirements Identification

The following methods were employed to identify and validate system requirements:

In order to identify the system requirements of the Remote Customer Onboarding System, a structured analysis process was followed.

First, the end-to-end onboarding process was examined step by step. The workflow of the system was analyzed from the moment a customer starts the application until the completion of account activation. This helped determine the main activities, decision points, and data exchanges within the system.

Second, the main stakeholders of the system were identified. These include the customer, bank employees, call center agents, and external verification services such as identity validation, SMS service, and biometric verification tools. By examining the interaction between these actors and the system, functional needs and system boundaries were clarified.

Third, the inputs, outputs, and data stores used during the onboarding journey were analyzed. Personal information, identity data, verification results, product preferences, and contract details were considered as critical data elements. This helped define what data the system should collect, process, store, and transfer.

Fourth, possible exceptions and alternative flows were considered. For example, failed SMS verification, unsuccessful NFC reading, selfie mismatch, or incomplete customer information were evaluated to understand how the system should respond to errors and incomplete processes.

Due to data privacy and confidentiality constraints, real banking data could not be included in this project. Instead, sample and synthetic documents were created to represent the system analysis process. These documents reflect the structure and logic of a real digital onboarding system and were used to identify system requirements and design system models.

The information below is based on benchmarks of digital customer onboarding processes in banks.

2.1 Process Steps

1. Customer starts onboarding
2. Customer enters identity information
3. System performs Mernis check
4. System sends SMS verification code
5. Customer verifies phone number
6. Customer uploads/scans ID
7. System performs ID and NFC verification
8. Customer selects product
9. Customer enters address and personal details
10. System performs selfie verification
11. System initiates call center verification
12. Customer defines password
13. System activates mobile banking
14. System generates and sends contract

2.2 System Input and Output Analysis

Inputs:

- Identity information
- Phone number
- OTP code
- ID data
- Address information
- Product selection
- Selfie image
- Password

Outputs:

- Verification results
- Error messages
- Application status
- Activation confirmation
- Contract document

2.3 Business Rule

- Customer identity must be verified via Mernis before proceeding
- SMS verification must be completed successfully
- ID and NFC validation must match identity data
- Biometric verification must be successful
- Call center approval is required for completion
- Password must meet security requirements
- Contract must be generated after successful onboarding

2.4 Stakeholder of Process**Internal Actors:**

- Digital Onboarding System
- Call Center Agent

External Actors:

- Customer
- Mernis System
- SMS Service Provider
- NFC/ID Verification Service
- Biometric Verification Service
- Mobile Banking System
- Bank Transfer Verification

3. Event Table

The event table below captures all significant events that the Remote Customer Onboarding System must respond to, including their triggers, sources, activities, system responses, and destinations.

#	Event Name	Trigger	Source	Activity	Response	Destination
1	Start Onboarding	User starts application	Customer	User fills initial application form	System initiates onboarding process	System
2	Mernis Check	Identity info entered	System	System sends identity data to Mernis service	Validate identity via Mernis	External System
3	SMS Verification	Phone number entered	System	System generates OTP and waits for input	Send OTP and verify	Customer
4	ID Verification	ID uploaded/scanned	Customer	User uploads/scans ID document	Validate ID information	System
5	Product Selection	User selects product	Customer	User browses and selects product	Save selected product	Database
6	Address Entry	Address entered	Customer	User enters address details	Store address info	Database
7	NFC Verification	NFC scan started	Customer	System reads NFC chip from ID	Read chip data	System
8	Selfie Check	Selfie captured	Customer	User captures selfie	Face verification	System
9	Call Center Verification	Video call started	System	Agent performs video identity check	Agent verifies identity	Call Center
10	Password Creation	User defines password	Customer	User creates password	Save credentials	Database
11	Mobile Activation	Activation triggered	System	System provisions mobile banking account	Activate mobile banking	System
12	Contract Completion	Process completed	System	System generates and finalizes contract	Send contract to customer	Customer

Table 1: Event Table

4. Entity-Relationship Diagram (ERD)

The diagram is organized into five logical zones: Address Hierarchy (purple), Customer (orange), Product (green), Application Core (blue), Verification (red), and Post-Application (dark green).

Customer and Address Side:

The Customer entity holds the customer's identity information (first name, last name, email/phone). Each customer's password is stored separately in PasswordCred with a 1:0..1 relationship, since a customer may not yet have set a password. The address hierarchy is normalized as a chain: Country → City → District → ZipCode, connected through 1:N relationships. CustomerAddress acts as a bridge table linking a customer to this hierarchy via zip_code, allowing a customer to have multiple addresses.

Product Side:

ProductType represents the higher-level category (e.g. deposit account, credit card), while Product is the specific banking product offered, connected to its category via type_id (1:N).

Application Core:

Application sits at the heart of the system. Every application is mandatorily linked to one Customer (1:N) and one Product (1:N). It carries application_id, customer_id, product_id, app_status, and created_at.

Verification Side:

Each application has 0..1 verification record (the Application → Verification relationship is 1:0..1). The main Verification table stores overall_status, and the six tables below it (MernisVerif, BiometricVerif, SMSVerif, CallCenterVerif, IDNFCVerif, BankTransferVerif) are each connected with a 0..1 relationship, meaning each verification method can be used at most once per application, or not at all.

Post-Application:

If an application is approved, a Contract is created (1:0..1), and afterward the contract is delivered via ContractDelivery (1:0..1). In parallel, MobileBankingAct represents mobile banking activation, linked to Application with 1:0..1.

Key Design Decisions:

Processes like verification, contracting, and activation are modeled as separate entities rather than attributes of Application because each has its own sub-types or delivery details. Normalizing the address hierarchy is a classic choice made for data consistency.

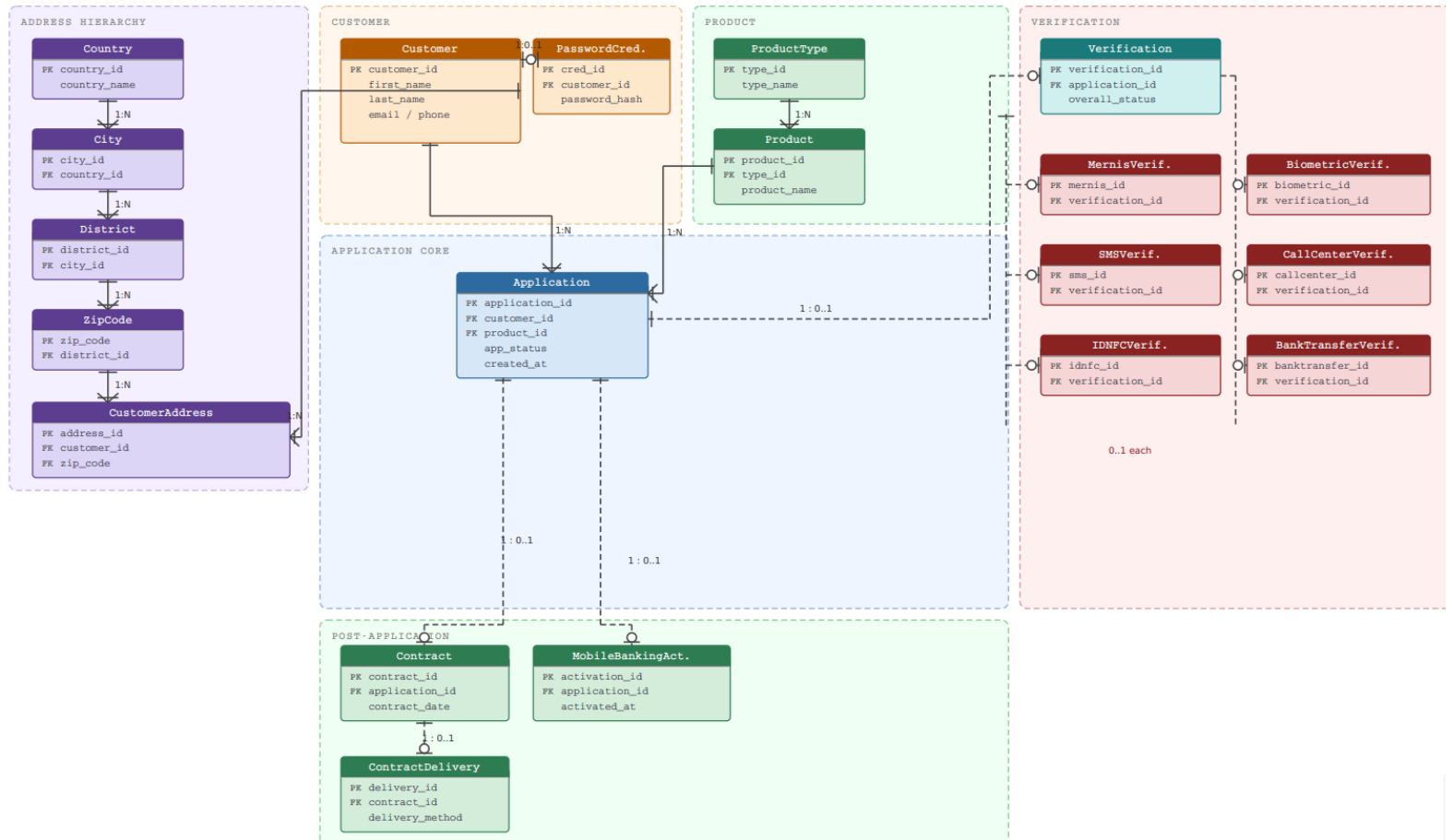


Figure 1: Entity Relationship Diagram (ERD)

5. Data Flow Diagrams (DFD)

5.1 Context Diagram

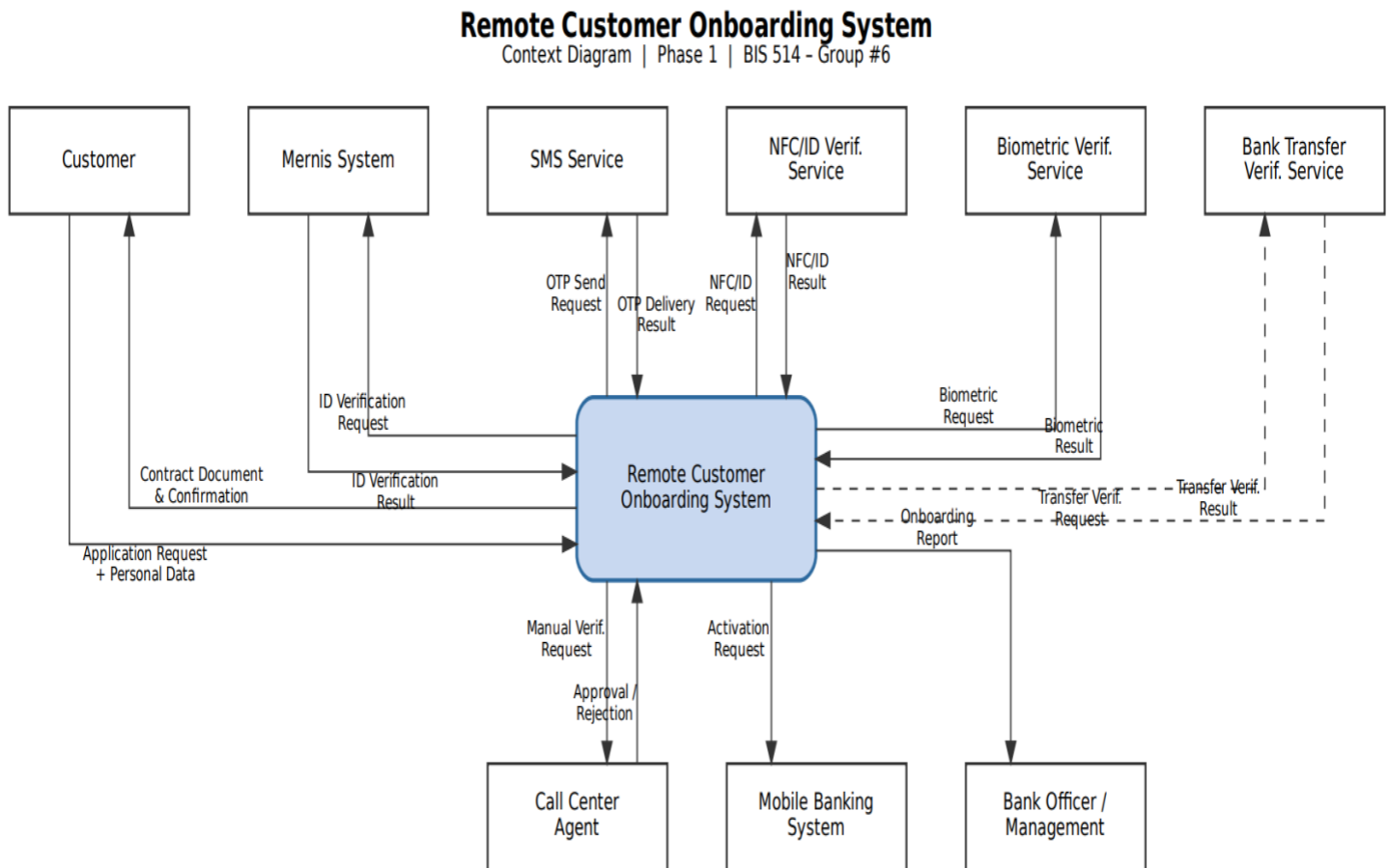


Figure 2: Context Diagram

5.2 0 Diagram

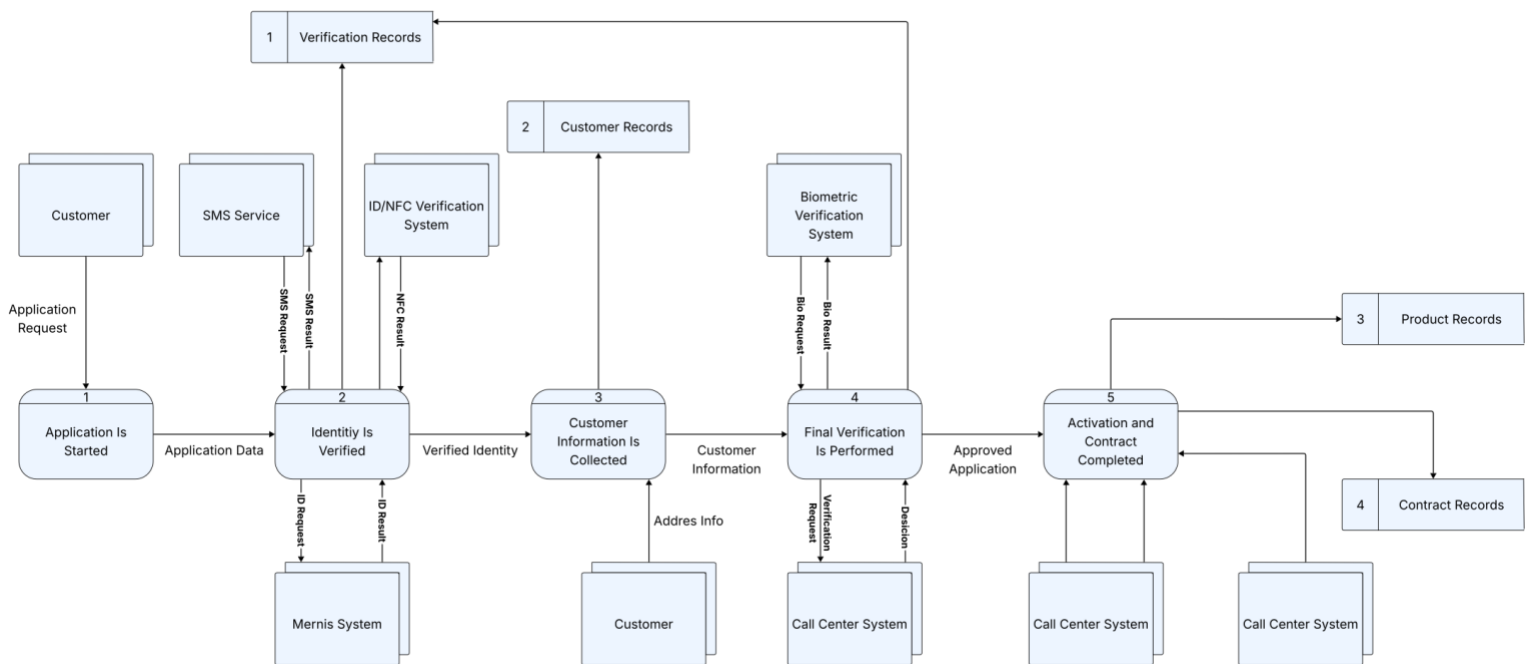


Figure 3: The Level 0 Diagram

5.3 Sub-diagram(s) Only for One Process of 0 Diagram

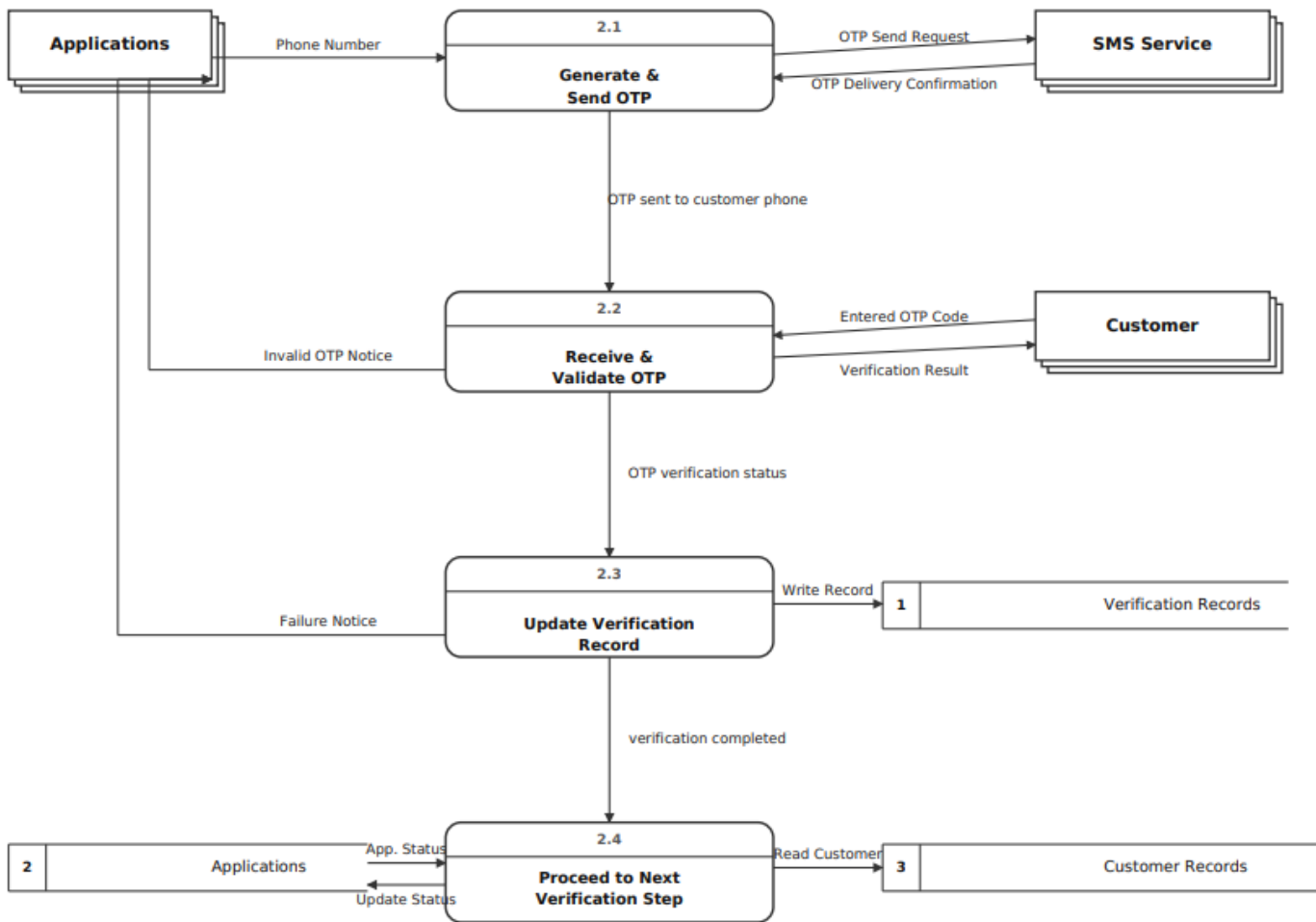


Figure 4: Sub Diagram of "Identity Is Verified"

5.3.2 Process Specification Form

PROCESS SPECIFICATION FORM
Number: 2
Name: Verify Phone Number via SMS
Description: This process verifies whether the phone number entered by the customer belongs to the applicant. The system sends a one-time password to the provided phone number through the SMS Service. The customer enters the received code, and the system checks whether the entered code matches the generated code. If the codes match, phone verification is marked as successful; otherwise, the process returns a failure result.
Input Data Flow: OTP Request Information Customer Phone Number Entered OTP Code
Output Data Flow: OTP Verification Result, Invalid OTP Notice, Verification Record Write, Update Status
Type of Process ☒ Online ☐ Batch ☐ Manual
Refer to Name: ☒ Structured English ☐ Decision Table ☐ Decision Tree
Unresolved Issues: Maximum retry count for incorrect OTP entries is not defined.

Table 2: Process Specification

Structured English:

RECEIVE Phone Number from Customer.

2.1 GENERATE & SEND OTP

CREATE OTP code.

SEND OTP Send Request to SMS Service.

RECEIVE OTP Delivery Confirmation from SMS Service.

SEND OTP to customer phone.

2.2 RECEIVE & VALIDATE OTP

RECEIVE Entered OTP Code from Customer.

VALIDATE OTP code.

IF OTP is valid THEN:

SEND Verification Result to Customer.

SET OTP verification status = SUCCESS.

ELSE (OTP is invalid):

SEND Invalid OTP Notice to Applications.

REDIRECT to Process 2.1 Generate & Send OTP.

ENDIF

2.3 UPDATE VERIFICATION RECORD

WRITE Verification Record to Verification Records (Record 1).

IF write is successful THEN:

SET verification status = completed.

SEND verification completed signal.

ELSE (write failed):

SEND Failure Notice to Applications.

ENDIF

2.4 PROCEED TO NEXT VERIFICATION STEP

READ Customer data from Customer Records (Record 3).

UPDATE App. Status in Applications (Record 2).

SEND Update Status to Applications.

PROCEED to Next Verification Step.

5.3.3 Data Flow Definition Form

DATA FLOW DESCRIPTION FORM	
Name: OTP Verification Status	
Description: This data flow carries the status of the phone number verification process after the entered OTP code is checked against the generated OTP.	
Source: 2.2 Receive & validate otp	Destination: 2.3 update verification record
Type of Data Flow: ☐ File ☐ Screen ☐ Report ☐ Form ☒ Internal	
Data Structure Traveling With the Flow: Verification Status + Phone Number + Verification Timestamp	Volume/Time: Generated once for each onboarding attempt after OTP validation.
Comments: This flow may contain either a successful or unsuccessful verification outcome.	

Table 3: Data Flow Description

5.3.4 Element Description Form

ELEMENT DESCRIPTION FORM													
Name: Verification Status													
Alias: OTP Status													
Alias: Phone Verification Result													
Description: This element indicates whether the phone number verification process was successful or unsuccessful.													
Element Characteristics Length: 10 Decimal Point: None Input Format: Text Output Format: Text Default Value: Pending	☐ Alphabetic ☒ Alphanumeric ☐ Date ☐ Numeric												
☐ Base or ☒ Derived													
Validation Criteria													
Continuous Upper Limit N/A Lower Limit N/A	<table border="0"> <tr> <td>Discrete</td> <td>Meaning</td> </tr> <tr> <td>Value</td> <td></td> </tr> <tr> <td>Verified</td> <td>OTP matched successfully</td> </tr> <tr> <td>Failed</td> <td>OTP did not match</td> </tr> <tr> <td>Pending</td> <td>Verification not completed</td> </tr> <tr> <td>yet</td> <td></td> </tr> </table>	Discrete	Meaning	Value		Verified	OTP matched successfully	Failed	OTP did not match	Pending	Verification not completed	yet	
Discrete	Meaning												
Value													
Verified	OTP matched successfully												
Failed	OTP did not match												
Pending	Verification not completed												
yet													
Comments This element is generated by the system after the OTP validation step is completed.													

Table 4: Element Description

5.3.5 External Agent Definition Form

EXTERNAL ENTITY DESCRIPTION FORM
Name: SMS Service
Description: The SMS Service is an external system used by the Digital Onboarding System to send one-time passwords to customers during phone number verification.
Input Data Flow: OTP Verification Request
Output Data Flow: OTP Delivery Status / OTP Verification Support Data
Notes: This entity is external because message delivery is handled outside the onboarding system.

Table 5: External Entity Description

5.3.6 Data Store Definition Form

DATA STORE DESCRIPTION FORM	
Id: D1	
Name: Verification Records	
Alias: Verification Data Store	
Description: This data store keeps the results of identity and verification-related checks performed during the digital onboarding process. It includes records for Mernis validation, SMS verification, ID/NFC verification, biometric verification, and final verification status.	
Data Store Characteristics File Type ☒ Computer ☐ Manual File Fomat ☒ Database ☐ Indexed ☐ Sequential Direct	
Recorde Size (Characters): Approx . 200	Average Number of Records: Depends on the number of onboarding applications Maximum Number of Records: (Secret Information) Percent Growth per Year: Expected to increase depending on customer onboarding volume (Secret Information)
Primary Key: Verification_ID Secondary Key: Application_ID	

Table 6: Data Store Description

6. Use Case Diagram

The Use Case Diagram identifies the actors interacting with the system and the primary use cases each actor participates in.

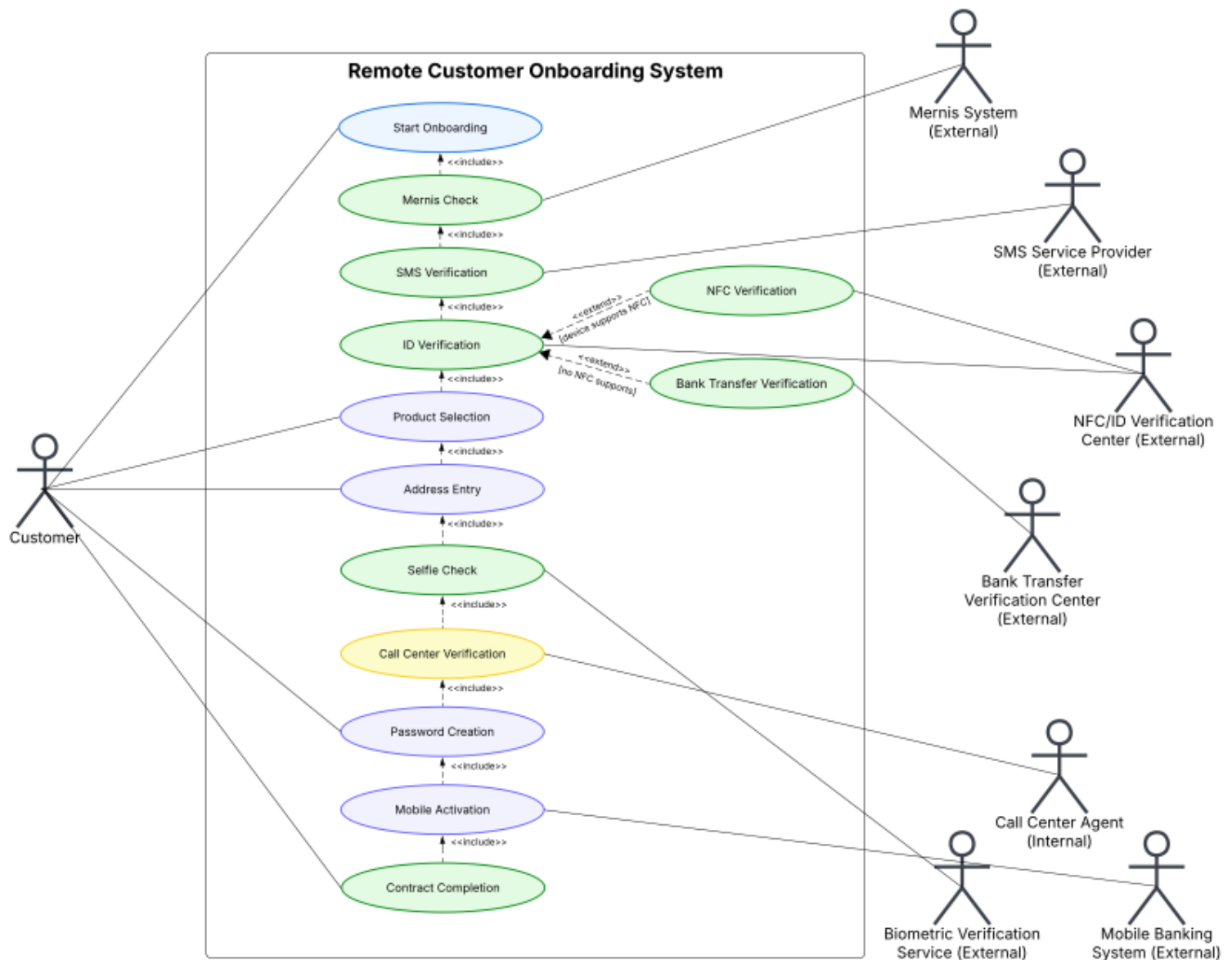


Figure 5: Use Case Diagram

7. Class Diagram (Without Attributes)

The Class Diagram represents the object-oriented view of the system. Attributes are excluded at this stage; only classes and their relationships are shown.

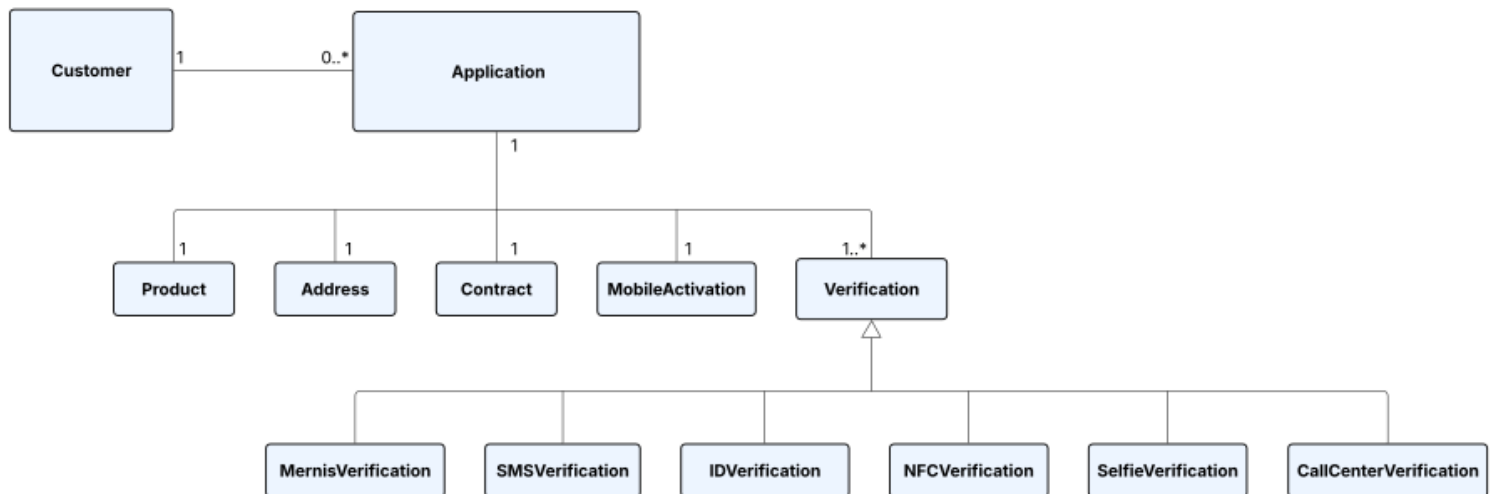


Figure 6: Class Diagram

8. Sequence Diagram

8.1 Scenario Description

Scenario: Successful Verification Process:

In this case, a customer uses the Remote Customer Onboarding System to successfully finish the verification procedure. When the customer starts the onboarding process, the system commences identity verification, which starts the verification process.

To confirm the customer's identity, the Verification System first transmits their identity details to the external Mernis System. After verifying the validity of the identity data, the Mernis System sends the outcome back to the Verification System.

The system then uses the SMS Service Provider to send a one-time password (OTP) to the customer's phone number. After receiving the SMS code, the client inputs it into the system. The SMS Service Provider receives the code from the Verification System and verifies its validity.

The NFC/ID Verification Center is then used by the system to verify identities. The system further completes NFC verification via the NFC/ID Verification Center as an optional step because the customer's device supports NFC. The NFC verification has been successfully finished. The system uses the Biometric Verification Service to take a selfie after the ID is verified.

The Biometric Verification Service receives the customer's selfie, examines it, and verifies that the liveness check was successful.

Call center verification is a required step in this system. Consequently, the Verification System notifies the Call Center Agent to conduct a manual verification once the selfie check is over. After reviewing the customer's data, the call center agent verifies that the verification process was successful. The customer is then informed by the Verification System that the verification process is complete and that all verification processes have been successfully performed.

8.2 Sequence Diagram

Remote Customer Onboarding System- Verification

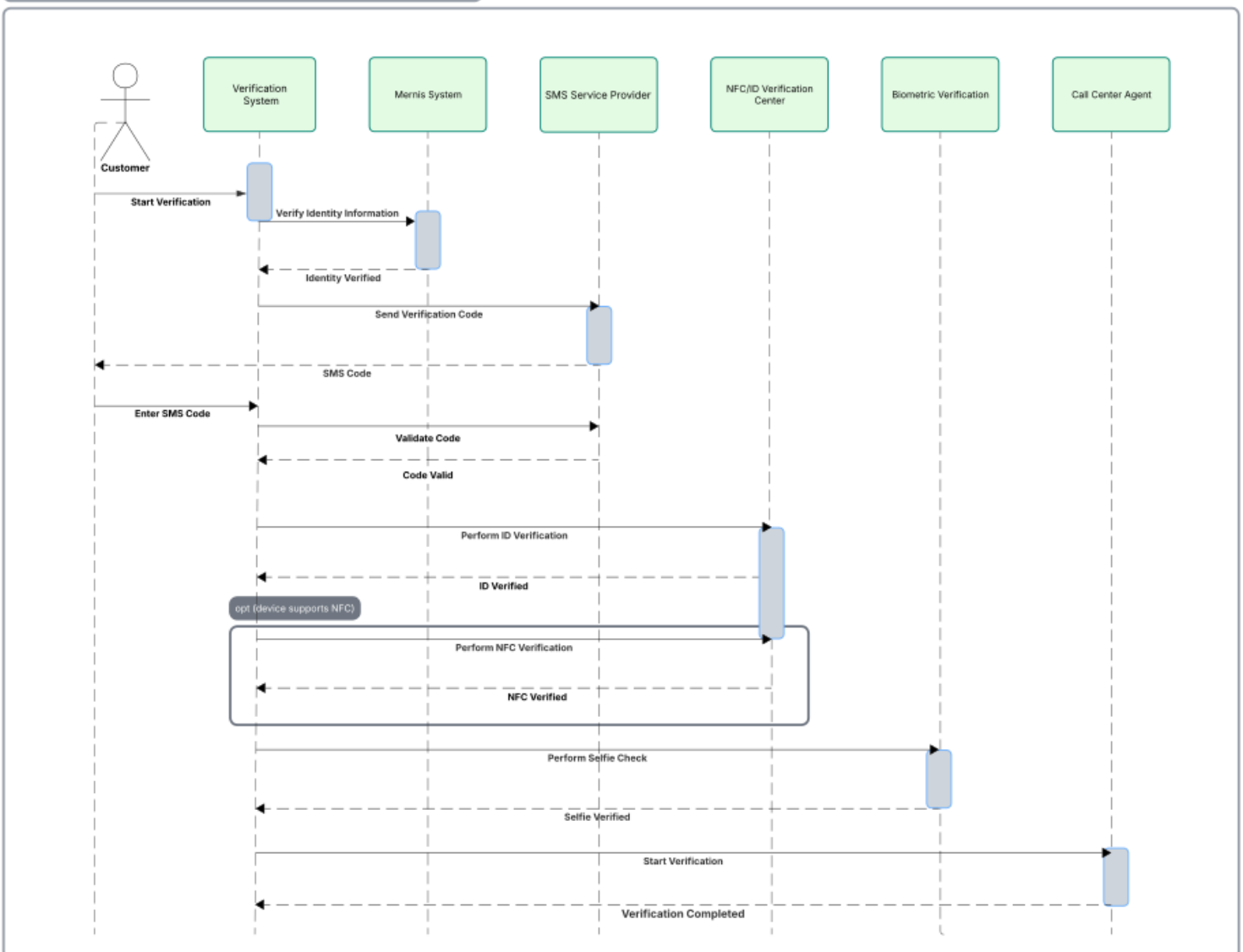


Figure 7: Sequence Diagram

9. Statechart Diagram

The Statechart Diagram below describes the lifecycle of the Verification class. It captures all possible states an application can be in and the transitions between them.

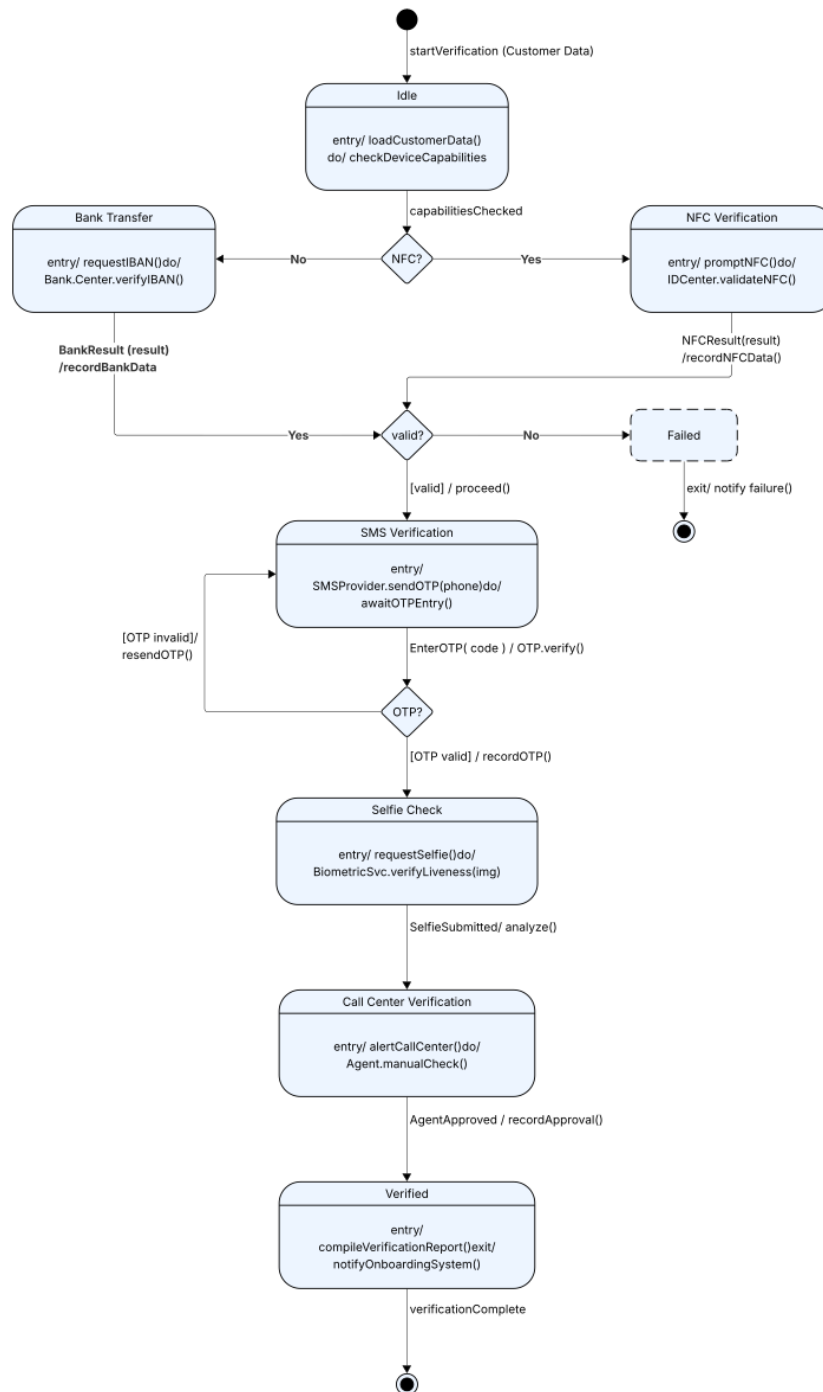


Figure 8: Statechart Diagram

10. Top Structure Chart

The top structure chart represents the highest-level decomposition of the Remote Customer Onboarding System. Top Level Structure Chart prepared by considering the 0 Level Data Flow Diagram. Each module corresponds to a major functional area of the system. The system is broken down into five main modules, each responsible for a distinct phase of the onboarding process.

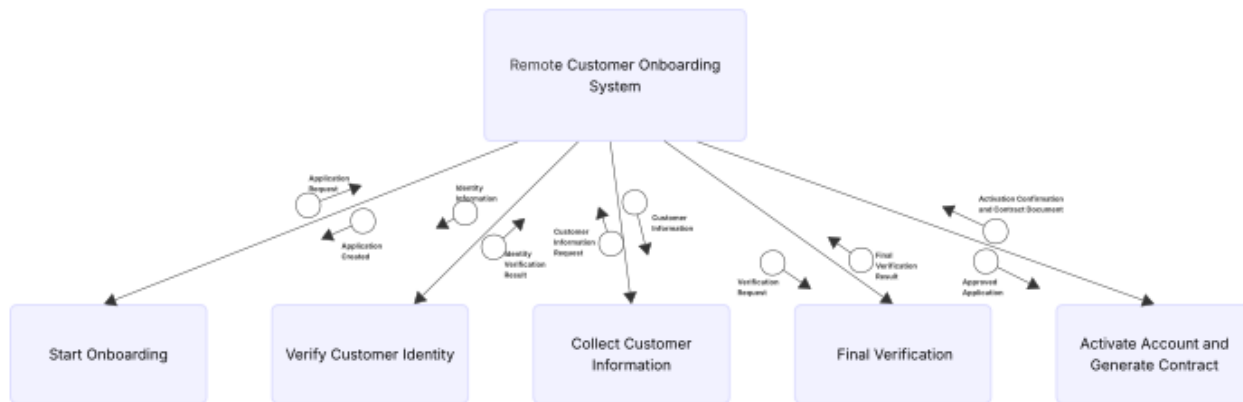


Figure 9: Top Structure Chart

10.1 Detailed Branch: “Verify Customer Identity”

The **Verify Customer Identity** module is selected for detailed decomposition as it is the most complex module with alternative paths (NFC vs. Bank Transfer). Below is the detailed structure chart of this branch.

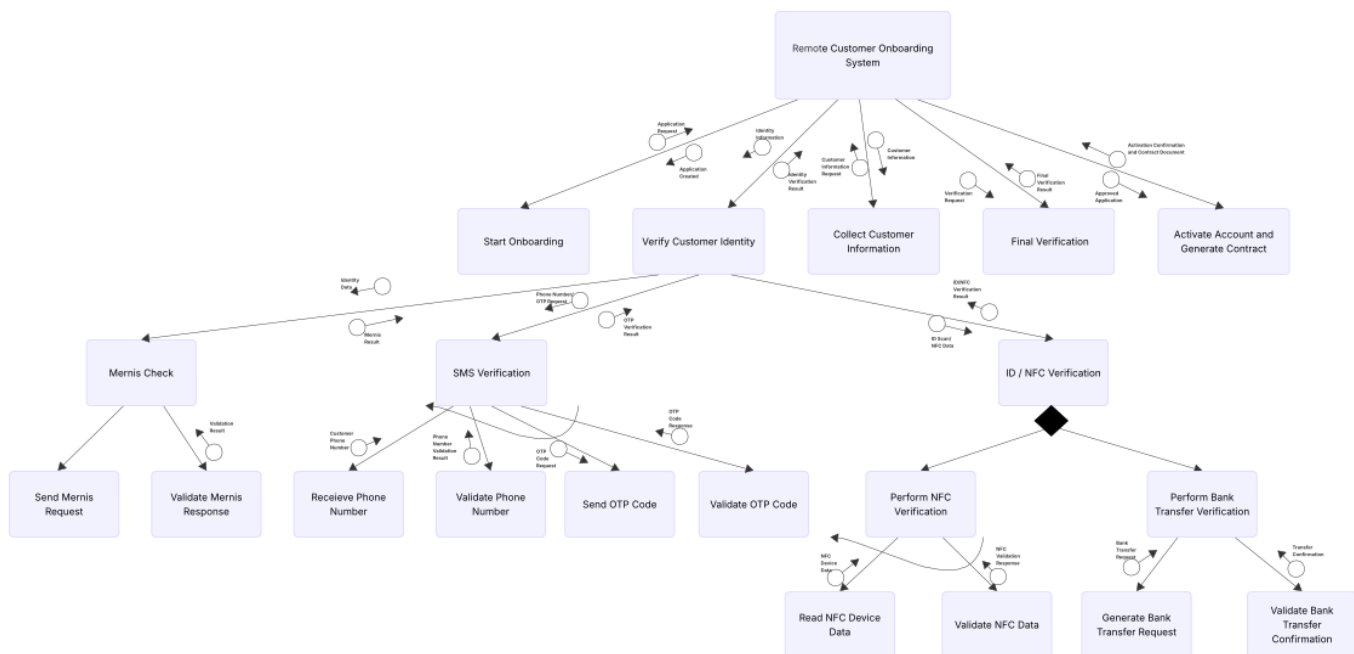
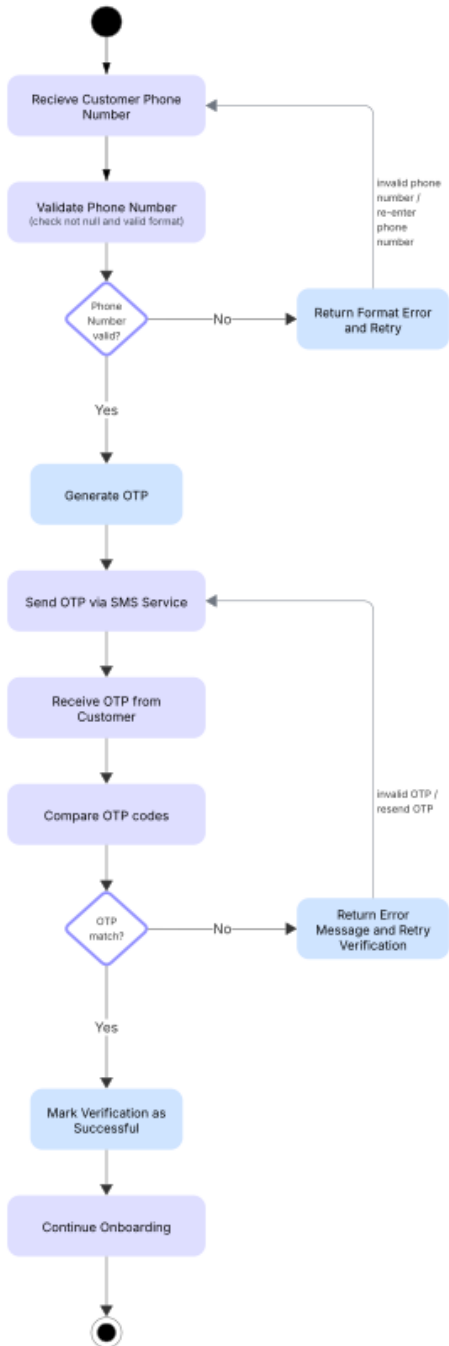


Figure 10: Detailed Branch of Top Structure Chart

10.2 Internal Logic for One Module: “SMS Verification”

The **SMS Verification** sub-module is responsible for obtaining the customer's phone number, sending an SMS code to that number, receiving the code back from the customer, and completing the SMS Verification step if it matches.



The SMS Verification module checks whether the phone number entered by the customer belongs to that person. The system asks the customer to enter a phone number; it must be valid, otherwise the system prompts the customer to enter the phone number again. It sends an OTP code via SMS service, then receives the code entered by the customer and compares it with the generated OTP. If the OTP codes match, the phone verification is successfully completed, and the registration process continues. If the codes do not match, the system returns a failed result and asks the customer to try again.

Figure 11: Internal Logic of SMS Verification

11. Package Diagram

The package diagram groups the design classes of the Remote Customer Onboarding System based on their responsibilities. The system consists of Customer, Application, Verification, Product, Contract, and Mobile Banking subsystems. Each subsystem contains related design classes, and dependency relationships illustrate the onboarding flow from customer application and verification to contract generation and mobile banking activation.

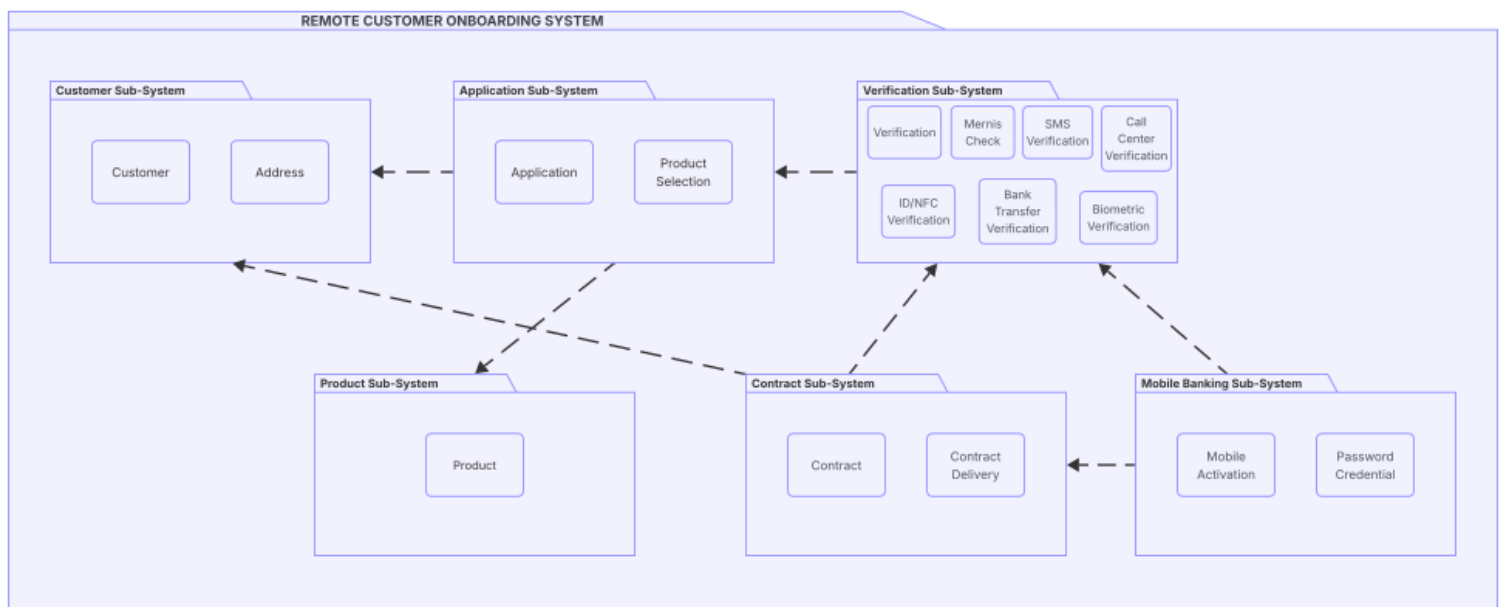


Figure 12: Package Diagram

12. Relational Database Design

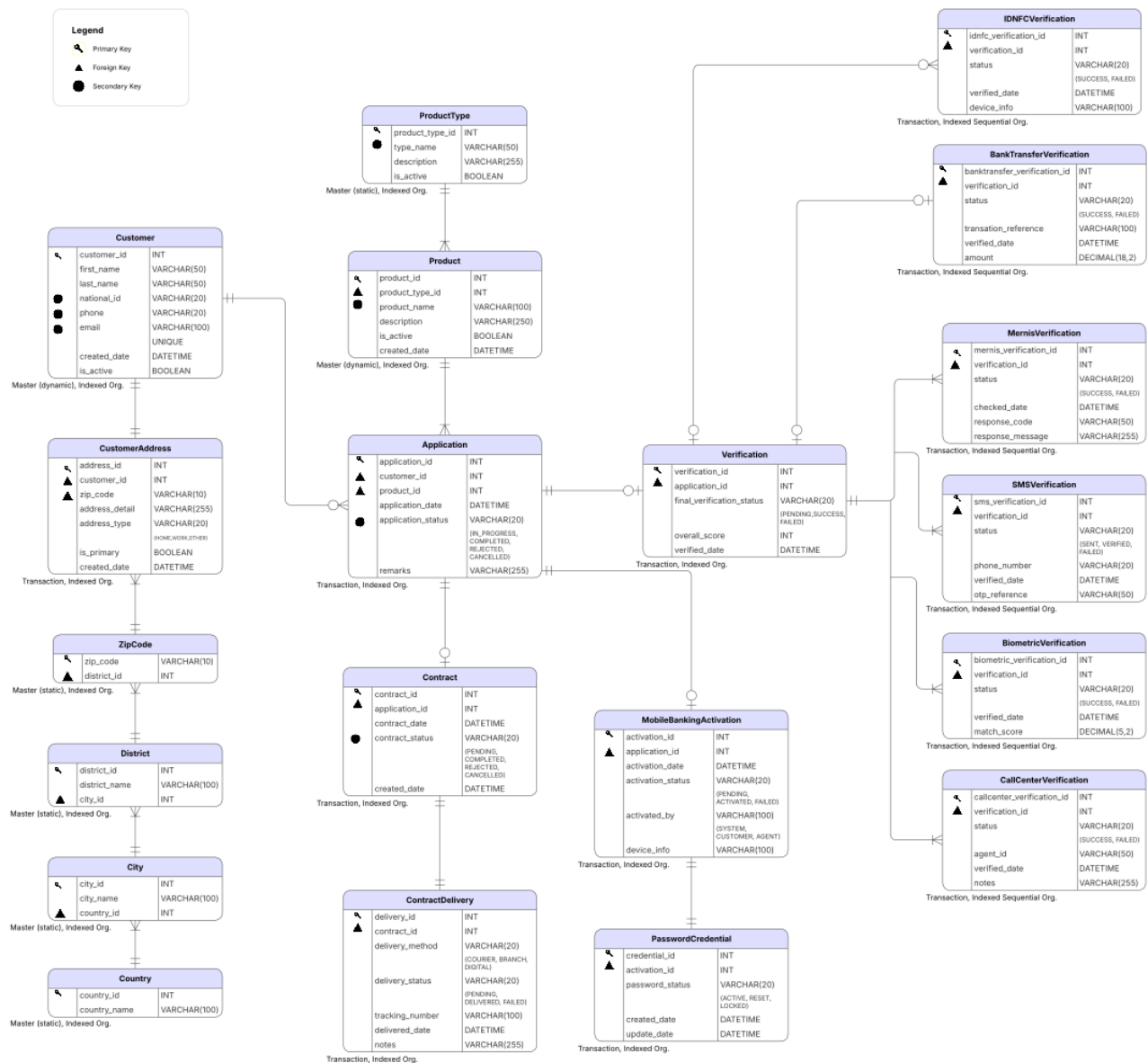


Figure 13: Relational Database Design

12.2 Relational Database Design System Explanation

Entity Relationships:

Customer - Application (One-to-Many)

One customer can start multiple onboarding applications, but each application belongs to only one customer.

Customer - CustomerAddress (One-to-One)

One customer must have one address, and each address belongs to only one customer.

Country - City (One-to-Many)

One country can contain multiple cities, while each city belongs to only one country.

City - District (One-to-Many)

One city can have multiple districts, but each district belongs to only one city.

District - ZipCode (One-to-Many)

One district may contain multiple zip codes, while each zip code belongs to one district.

ZipCode - CustomerAddress (One-to-Many)

One zip code can be used in multiple customer addresses, while each address is linked to one zip code.

ProductType - Product (One-to-Many)

One product type can include multiple products, but each product belongs to only one product type.

Product - Application (One-to-Many)

One product can be selected in multiple applications, while each application is linked to one selected product.

Application - Verification (One-to-One / Optional)

Each application may have one verification record, but a verification record belongs to only one application.

Verification - MernisVerification (One-to-Many)

A verification process includes Mernis check, and each Mernis verification record belongs to one verification process.

Verification - SMSVerification (One-to-Many)

A verification process includes SMS verification, and each SMS verification record belongs to one verification process.

Verification - IDNFCVerification (One-to-One / Optional)

A verification process may include ID/NFC verification, and each ID/NFC verification record belongs to one or zero verification process.

Verification - BiometricVerification (One-to-Many)

A verification process includes biometric verification, and each biometric verification record belongs to one verification process.

Verification - CallCenterVerification (One-to-Many)

A verification process includes call center verification, and each call center verification record belongs to one verification process.

Verification - BankTransferVerification (One-to-One / Optional)

A verification process may include one bank transfer verification if ID/NFC verification cannot be completed, and each bank transfer verification record belongs to one verification process.

Application - Contract (One-to-One / Optional)

An approved application may generate one contract record, but each contract belongs to only one application.

Contract - ContractDelivery (One-to-One / Optional)

A contract may have one delivery record that stores how the contract is delivered to the customer.

Application - MobileBankingActivation (One-to-One / Optional)

An approved application may result in one mobile banking activation record, and each activation belongs to one application.

MobileBankingActivation – PasswordCredential (One-to-One)

A completed mobile banking activation generates one password credential, and each password credential belongs to one mobile banking activation.

Note: ID/NFC Verification and Bank Transfer Verification are alternative identity verification methods. For a single Verification record, only one of these two methods should be completed.

13. Design of System Outputs

The following list identifies all the necessary outputs generated by the Remote Customer Onboarding System. Each output supports a specific stakeholder, ranging from customers (receiving confirmations) to bank officers (receiving onboarding analytics). Detailed designs are provided for three representative outputs.

List of System Outputs:

1. Application Confirmation Notice — confirms the application has been registered.
2. OTP / SMS Verification Result — shows whether SMS verification succeeded or failed.
3. Identity Verification Summary — consolidated status of Mernis, NFC/ID and biometric checks.
4. Contract Document — final onboarding contract delivered to the customer.
5. Mobile Banking Activation Confirmation — confirms successful activation of mobile banking.
6. Onboarding Report (for Bank Officer) — aggregated daily/weekly onboarding statistics.
7. Rejection / Failure Notice — informs the customer when onboarding cannot be completed.

13.1 Application Confirmation Notice

Output Type: Detailed Notice (Email / In-App)

Purpose: Confirms that the customer's onboarding application has been successfully received and registered.

Security Controls: Sent only to the registered customer email/phone. No sensitive data exposed.

Design:

APPLICATION CONFIRMATION NOTICE	
Notice Date	14.05.2026
Application ID	APP-100245
Customer Name	Fatih Özdal
Selected Product	Savings Account
Status	Registered – Awaiting Verification
Next Step	SMS Verification

13.2 Identity Verification Summary

Output Type: Detailed Report (Internal / Customer-Facing Summary)

Purpose: Summarizes the result of all identity verification steps performed during onboarding.

Security Controls: Accessible only by authenticated users; encrypted in transit and at rest.

Design:

IDENTITY VERIFICATION SUMMARY			
Application ID	APP-100245		
Customer Name	Fatih Özdal		
Verification Step	Status	Timestamp	Source
Mernis Check	Successful	14:22	Mernis System
SMS Verification	Successful	14:24	SMS Service
ID / NFC Verification	Successful	14:27	NFC/ID Service
Biometric (Selfie)	Successful	14:30	Biometric Svc.
Call Center Approval	Approved	14:38	Call Center Agent
Overall Status	VERIFIED – READY FOR ACTIVATION		

14. Design of System Inputs

The following list identifies all the necessary inputs to the Remote Customer Onboarding System. Each input feeds a specific process and is subject to validation rules. Detailed designs are provided for three representative inputs.

List of System Inputs:

1. Application Registration Input — initial personal data submission.
2. OTP Code Input — code entered to confirm phone ownership.
3. ID Document Upload / NFC Read Input — identity document scan or NFC chip data.
4. Selfie / Biometric Capture Input — live photo for face matching.
5. Address Information Input — residential and mailing address details.
6. Product Selection Input — chosen banking product (savings, credit card, etc.).
7. Password Definition Input — credential for mobile banking activation.

14.1 Application Registration Input

Purpose: To register a new customer and initiate the onboarding process.

Data Fields:

Field	Type	Validation
First Name	Text	Cannot be empty
Last Name	Text	Cannot be empty
National ID Number	Numeric	Must be 11 digits; Mernis-validated
Date of Birth	Date	Customer must be 18 years or older
Email	Email	Valid format; must be unique
Phone Number	Numeric	Valid country format; must be unique

Input Controls: Required-field validation; Mernis cross-check; email/phone uniqueness; rate limiting.

Input Device: Mobile Smartphone, Tablet, Desktop / Laptop.

Code Type Used: Simple Sequence Code for Application_ID.

Design:

20:31

Kişisel Bilgiler

1 - 2 - 3 - 4 - 5 - 6

T.C. Kimlik No Giriniz

Doğum Tarihi GG.AA.YYYY

E-posta İsteğe Bağlı

Ülke Kodu +90 (Türkiye) >

Telefon No (5xx) xxx xx xx

Girdiğim cep telefonu numarası kolay adres olarak tanımlansın. ☐

Kolay adres tanımlaması yaparak, para transferlerinizde IBAN yerine cep telefonu numaranızı paylaşabilirsiniz.

Davet Kodu İsteğe Bağlı

Davet kodu, yakınlarınızın size ilettiği ya da Kuveyt Türk'ün sunduğu size özel avantajlar sağlayan kampanya kodudur.

Açık Rıza Onayı

Biyometrik Veri Açık Rıza Metni

Kişisel Verilerle İlgili

14.2 OTP Code Input

Purpose: To confirm that the customer is the owner of the registered phone number.

Data Fields:

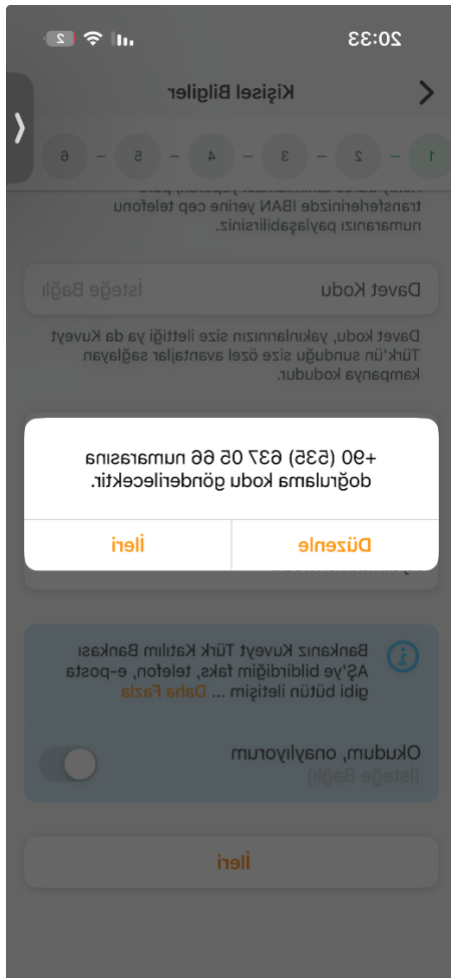
Field	Type	Validation
OTP Code	Numeric	Exactly 6 digits
Phone Number (auto)	Numeric	Pre-filled from session
Submission Timestamp	DateTime	Auto-captured; must be within 180 sec of OTP generation

Input Controls: Format check, expiration check, maximum 3 retry attempts.

Input Device: Mobile Smartphone, Tablet, Desktop / Laptop.

Code Type Used: Simple Sequence Code for SMS_Verification_ID.

Design:



14.3 Selfie / Biometric Capture Input

Purpose: To verify that the customer matches the photo on their identity document.

Data Fields:

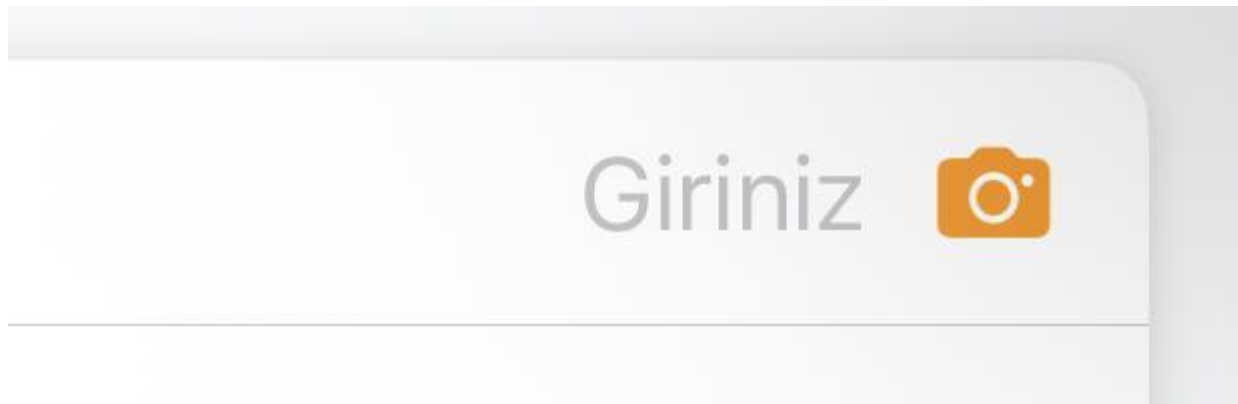
Field	Type	Validation
Selfie Image	Image (JPG/PNG)	Min 480p, max 5 MB, single face detected
Liveness Check Token	Alphanumeric	Generated during capture; required
Capture Timestamp	DateTime	Auto-captured; must be live (no upload)

Input Controls: Liveness detection, file size validation, face match score ≥ 0.85 .

Input Device: Mobile Smartphone (front camera), Tablet.

Code Type Used: Simple Sequence Code for Biometric_Verification_ID.

Design:



15. Menu Hierarchy

The menu hierarchy of the Remote Customer Onboarding System maps the back-end subsystems to the customer- and operator-facing navigation. It provides intuitive access for customers progressing through onboarding and for bank staff overseeing the process.

1. Customer Portal

- Start New Application
- Resume Application
- Application Status Tracking
- Document Download (Contract)

2. Verification Subsystem

- Mernis Identity Check
- SMS / OTP Verification
- ID / NFC Verification
- Biometric (Selfie) Verification
- Call Center Verification

3. Product & Application Management

- Browse Banking Products
- Select Product
- Application History

4. Address & Profile Management

- Add / Edit Address
- Contact Information
- Password & Security Settings

5. Contract & Activation

- View Contract
- Contract Delivery Method
- Mobile Banking Activation

6. Bank Officer / Admin Panel

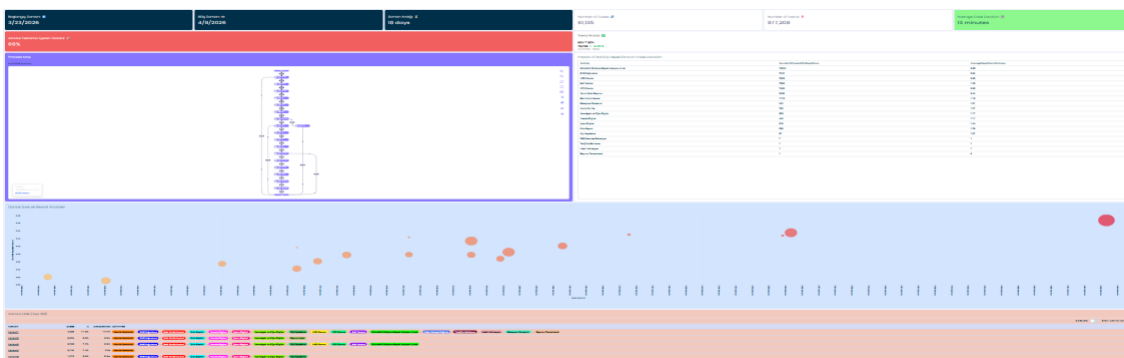
- Application Review
- Verification Queue
- Onboarding Reports
- Failed Applications

7. Support & Help

- FAQ
- Live Chat
- Call Center Contact
- Language Settings

Design:

Bank Officer/Admin Panel



16. Test Case (SMS Verification with Invalid OTP)

The following test case verifies that the system correctly rejects an incorrect OTP code entered during phone verification.

positive

Test Scenario Code and Name	SMS – OTP Verification
Test Case Code and Name	SMS - TC-01 - Entering an OTP code that does not match the system-generated OTP
Preconditions	1. The customer has started a new onboarding application. 2. Mernis identity check has been completed successfully. 3. An OTP code has been sent to the customer's registered phone number. 4. The OTP has not yet expired (within 180 seconds).
Steps	1. The system displays a text box for entering the OTP code. 2. The customer enters a 6-digit code. 3. The customer clicks the "Verify" button.
Test Data	1. System-generated OTP: "482915" 2. Customer-entered OTP: "123456" 3. Click the "Verify" button
Expected Result	System displays the error message "The OTP code you entered is incorrect. Please try again." The verification status is set to "Failed", and the customer is prompted to retry (up to 3 attempts).
Result	Failed -The system rejected the invalid OTP but customer failed.

Negative

Test Scenario Code and Name	SMS – OTP Verification
Test Case Code and Name	SMS - TC-01 - Entering an OTP code that does not match the system-generated OTP
Preconditions	1. The customer has started a new onboarding application. 2. Mernis identity check has been completed successfully. 3. An OTP code has been sent to the customer's registered phone number. 4. The OTP has not yet expired (within 180 seconds).
Steps	1. The system displays a text box for entering the OTP code. 2. The customer enters a incorrect 6-digit code. 3. The customer clicks the "Verify" button.
Test Data	1. System-generated OTP: "482915" 2. Customer-entered OTP: "123456" 3. Click the "Verify" button
Expected Result	System displays the error message "The OTP code you entered is incorrect. Please try again." The verification status is set to "Failed", and the customer is prompted to retry (up to 3 attempts).
Result	Passed -The system rejected the invalid OTP but customer stil passed.

17. Appendix

This appendix documents the requirement elicitation activities used to support the analysis of the Remote Customer Onboarding System. The section includes a sample questionnaire, summarized questionnaire findings translated into system requirements, and document analysis/benchmarking results from comparable digital banking onboarding practices.

Appendix A: Requirements Elicitation Questionnaire

Target Audience: 50 individuals aged 18 and above who have experience using mobile banking or opening financial accounts through digital channels.

Objective: To identify user expectations, pain points, trust factors, and functional requirements for a secure and user-friendly remote customer onboarding system.

Part 1: Digital Banking Usage & Onboarding Expectations

1. How often do you use mobile or internet banking services?

- ☐ Frequently (Several times a week)
- ☐ Occasionally (A few times a month)
- ☐ Rarely
- ☐ Never

2. What is your main concern when becoming a bank customer remotely? (Select top two)

- ☐ Security of personal data
- ☐ Accuracy of identity verification
- ☐ Difficulty in using NFC or ID scanning
- ☐ Long waiting time for video call verification
- ☐ Unclear error messages

3. Which verification step do you think is the most critical in remote onboarding?

- ☐ Mernis identity check
- ☐ SMS/OTP verification
- ☐ ID card and NFC verification
- ☐ Selfie/liveness verification
- ☐ Call center/video call approval

4. What would make you trust a remote onboarding process more?

- ☐ Clear step-by-step guidance
- ☐ Real-time application status tracking
- ☐ Visible security and privacy explanations
- ☐ Fast support from call center
- ☐ Confirmation messages after each step

Part 2: User Experience & Process Efficiency

5. Which device would you primarily use to complete remote customer onboarding?

- ☐ Mobile smartphone
- ☐ Tablet
- ☐ Desktop/Laptop

6. Which feature is a must-have during remote onboarding?

- ☐ Save and continue later
- ☐ Application status tracking
- ☐ Live support/chat assistance
- ☐ Clear retry options after failed verification
- ☐ Short estimated completion time

7. If NFC or ID scanning fails, what should the system provide?

- ☐ Detailed retry instructions
- ☐ Alternative verification method
- ☐ Call center support option
- ☐ Option to restart from the failed step only

8. How should the contract be delivered after successful onboarding?

- ☐ Email
- ☐ SMS link
- ☐ Mobile banking notification
- ☐ Downloadable PDF within the application

9. Which notification channel do you prefer during onboarding?

- ☐ SMS
- ☐ Email
- ☐ Mobile push notification
- ☐ In-app notification

10. How important is it to see the current status of your application?

- ☐ Extremely important
- ☐ Somewhat important
- ☐ Not important

Appendix B: Summary of Questionnaire Results & Derived System Requirements

Based on the responses from 50 participants, the following findings were gathered and translated into functional requirements for the Remote Customer Onboarding System.

Finding	Questionnaire Result	Derived System Requirement
Finding 1: Security and privacy concerns	82% of respondents selected personal data security and secure identity verification as their primary concern.	SR-01: The system MUST display clear security and privacy information before collecting personal or biometric data. SR-02: The system MUST store verification records securely with timestamped audit information.
Finding 2: Step-by-step process guidance	76% of respondents stated that clear guidance and confirmation after each step would increase trust.	SR-03: The system MUST provide a step-by-step progress indicator showing the current onboarding stage. SR-04: The system MUST show success or failure messages after each verification step.
Finding 3: NFC and ID scanning difficulties	64% of respondents considered NFC/ID scanning as the most likely point of failure.	SR-05: The system MUST provide retry instructions for failed NFC or ID verification. SR-06: The system SHOULD support an alternative verification route such as bank transfer verification or call center assistance.
Finding 4: Application status tracking	70% of respondents indicated that seeing the current application status is extremely important.	SR-07: The system MUST provide real-time application status tracking, such as Started, Mernis Verified, SMS Verified, ID/NFC Verified, Call Center Pending, Approved, or Rejected.
Finding 5: Mobile-first onboarding	68% of respondents preferred using a mobile smartphone for remote onboarding.	SR-08: The system MUST be responsive and optimized for mobile devices, especially for ID scanning, NFC reading, selfie capture, and video call verification.
Finding 6: Contract and activation confirmation	60% of respondents preferred receiving the contract through email or a downloadable PDF after successful onboarding.	SR-09: The system MUST generate the contract after approval and deliver it through email or an in-app downloadable document. SR-10: The system MUST send activation confirmation once mobile banking is activated.

Appendix C: Document Analysis & Competitor Benchmarking

Objective: To establish practical standards for secure remote banking onboarding, focusing on identity verification, user experience, error handling, application tracking, and regulatory compliance.

Methodology: A document analysis and benchmarking approach was used. Publicly observable remote onboarding flows and general digital banking practices were reviewed to identify common standards and translate them into system requirements. Because real banking data and confidential internal documentation cannot be used, the analysis relies on synthetic process documentation and general industry practices.

Platforms / Practices Reviewed:

Digital bank mobile onboarding flows that use identity information, SMS verification, ID card scanning, NFC reading, selfie/liveness verification, and video call confirmation.

Mobile banking activation flows that deliver contracts electronically and activate the customer after successful verification.

Banking onboarding support practices that provide retry options, call center support, and status notifications when verification fails or remains incomplete.

Key Findings & Established Standards:

Benchmark Area	Observation	Derived System Requirement
Identity Verification Flow	Remote onboarding generally requires several sequential controls such as identity information entry, official identity validation, SMS verification, ID/NFC verification, biometric verification, and final approval.	SR-11: The system MUST enforce sequential verification rules and must not allow the customer to proceed if a mandatory verification step fails.
Error Handling and Alternative Paths	Critical friction points occur when SMS, NFC, selfie, or video call verification fails. A good onboarding flow should not force the customer to restart the whole process unnecessarily.	SR-12: The system MUST return the customer to the failed step with a clear explanation and retry option. SR-13: The system SHOULD support alternative verification where legally and operationally possible.
User Trust and Transparency	Customers are more likely to complete the process when they understand why data is collected and what step they are currently completing.	SR-14: The system MUST show short explanations for sensitive data collection steps such as NFC, selfie, and video call verification.
Application Tracking and Notifications	Modern digital onboarding systems provide status updates and confirmation messages to reduce uncertainty.	SR-15: The system MUST notify the customer after major milestones such as SMS verification, call center approval, mobile activation, and contract delivery.

Compliance and Auditability

Banking onboarding must be traceable. Verification attempts, results, timestamps, and final status should be stored for control and audit purposes.

SR-16: The system **MUST** maintain audit-ready verification records linked to the Application_ID and Verification_ID.

Appendix Conclusion

The appendix supports the requirements identification section by showing how user expectations and benchmarked digital banking practices were converted into system requirements. The results emphasize security, mobile usability, clear process guidance, controlled retry mechanisms, application status tracking, and reliable contract delivery as critical components of the Remote Customer Onboarding System.